

Quark/gluon jet discrimination in dijet photoproduction at the EIC

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Gluon jets are wider than quark jets, and quark/gluon tagging exploits that. We compare three observables in dijet photoproduction: the integrated jet shape $\Psi(r = 0.3)$, which measures how a jet's transverse energy is spread in radius; subjet multiplicity n_{subjets} , and the iterated soft-drop multiplicity n_{SD} , which counts splittings passing a momentum-fraction cut. All three are computed on exactly the same jets, from PYTHIA 8 events at $\sqrt{s} = 64, 105, 141$ and 300 GeV, covering the three EIC design points and a HERA reference.

Even though there is an increasing preference for machine learning techniques, theoretically grounded methods remain crucial for improving the interpretability of results and addressing biases in the simulated or experimental data used for training. There are many other complex observables that can be used to tag the jet based on the same physical content, expressed in different functional forms. This paper presents the first study of soft-drop multiplicity in dijet photoproduction events and compares it with the integrated jet shape and subjet multiplicity.